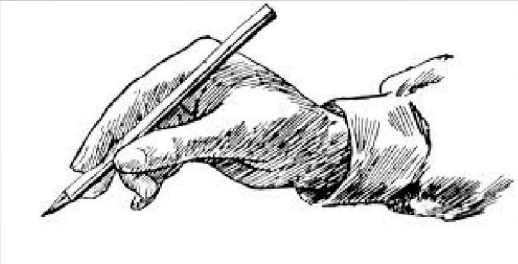
DEXTERITY CHECK

APMAE4990: INTRODUCTION TO DATA SCIENCE

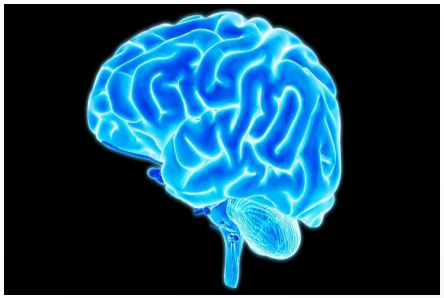
SPRING 2018

WHAT IS DEXTERITY CHECK?

- Dexterity check is a tool that measures hand motor function by asking the subject to submit drawings of a ‘circle’ ‘triangle’ ‘heart’ and ‘star’
- Then four different algorithms are used to determine what shape you drew
- Your performance is measured simply as “scores” of the number of correctly identified shapes.

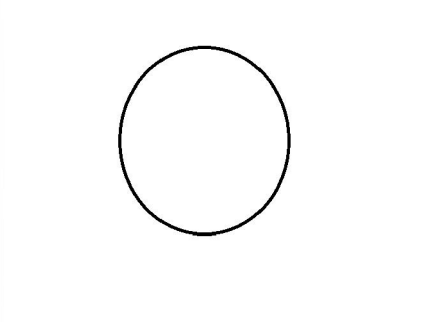


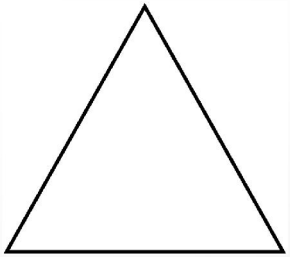
MOTIVATION

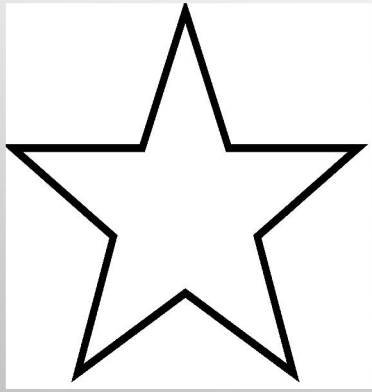


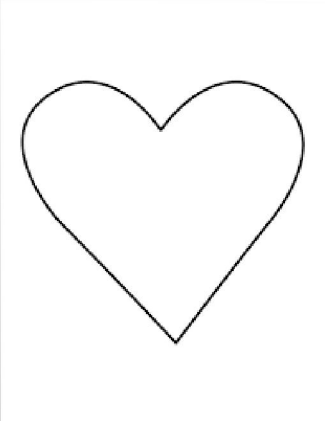
- Every year approximately 800,000 people suffer from stroke which causes impairment of upper extremity functioning
- Reaching and grasping studies are performed to study hand and arm impairment
- Movement disorders are usually diagnosed with a neurological exam, MRI scan, or an electromyography.

THE SHAPES

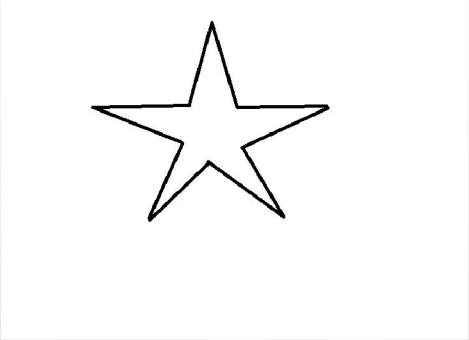


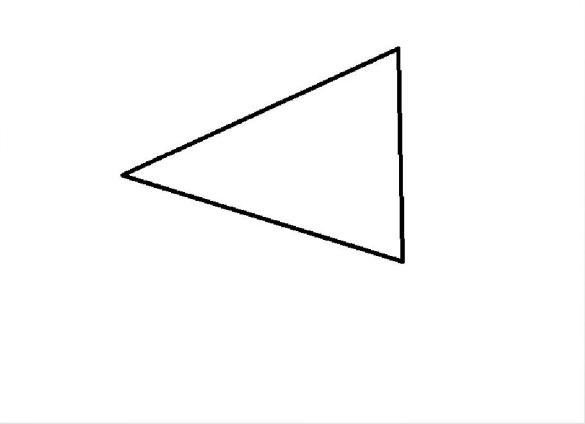


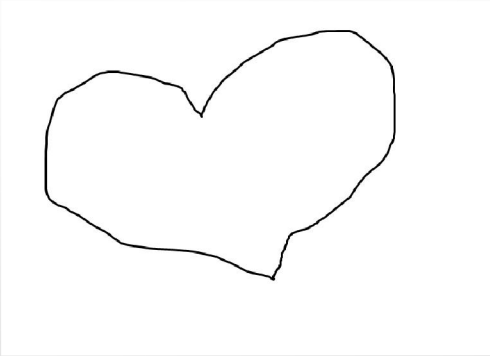


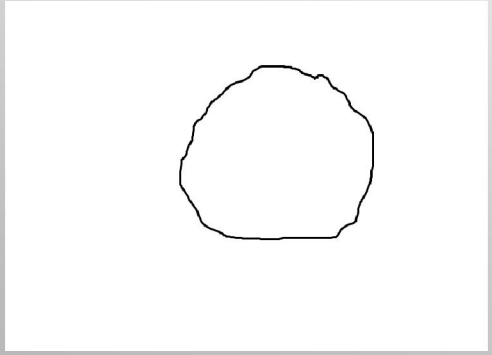


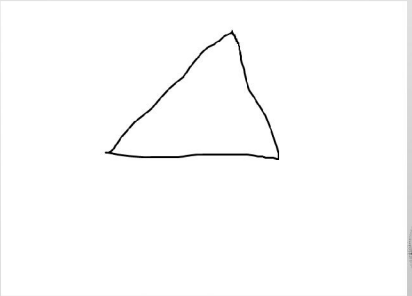
SUPERVISED PROBLEM: MIXED TRAINING SET





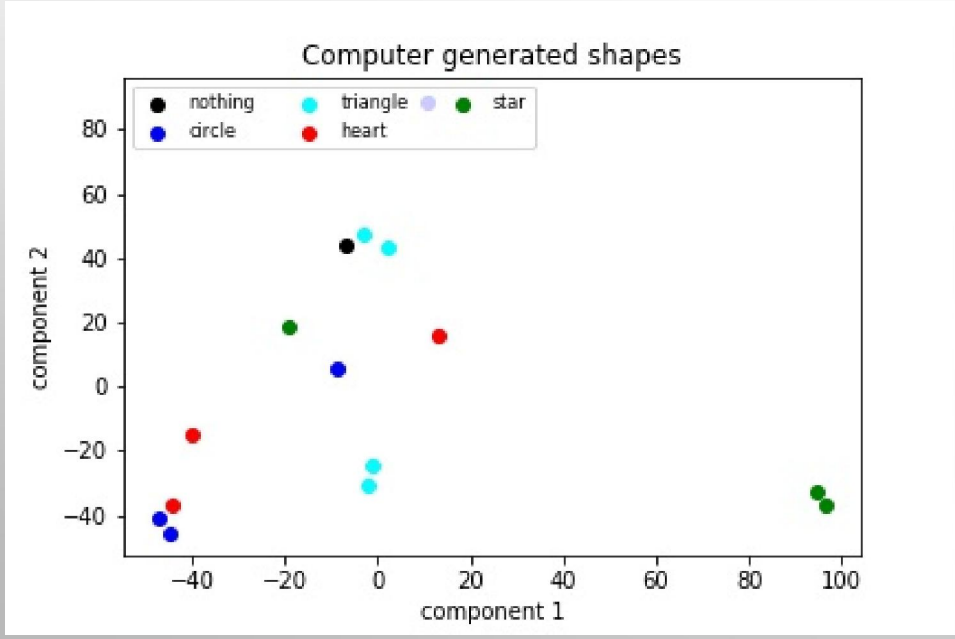




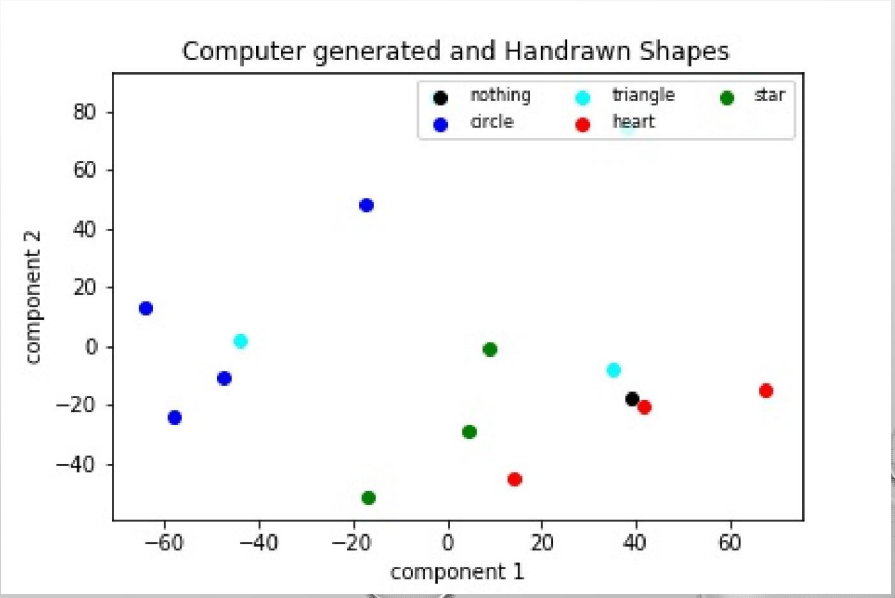


PCA COMPARISON OF MIXED TRAINING SET VERSUS COMPUTER GENERATED ONLY

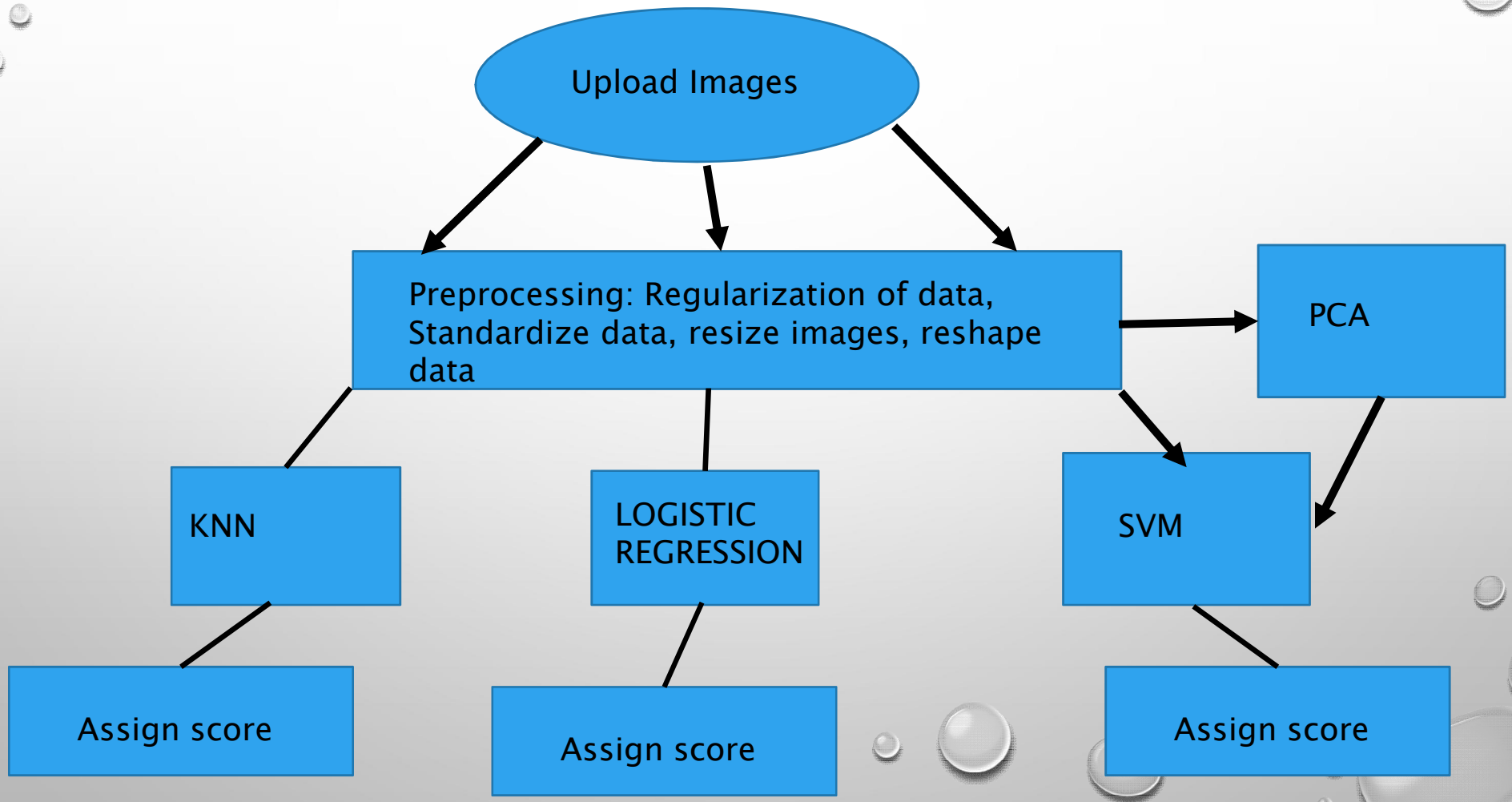
Computer Generated Only data set



Computer Generated and Hand drawn data set (Mixed)



FLOWCHART


Upload Images
Preprocessing: Regularization of data, Standardize data, resize images, reshape data
PCA
KNN
LOGISTIC REGRESSION
SVM
Assign score
Assign score
Assign score

MULTI CLASS CLASSIFICATION METHODS:

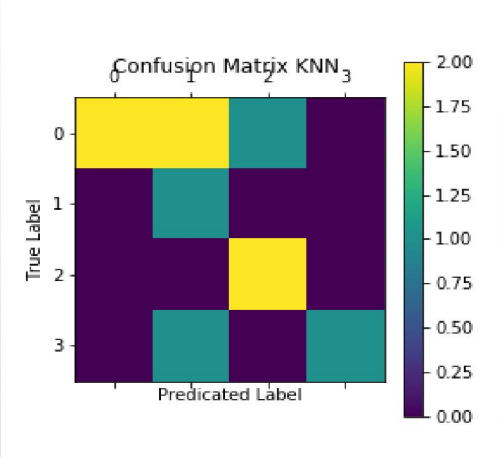
- 80% TRAINING SET 20% TEST SET WITH RANDOMIZATION
- GRIDSEARCH CV TO FIND OPTIMAL ‘K’ OR ‘ALPHA’ WITH CV=5
- L2 REGULARIZATION FOR HIGHER ACCURACY
- CONFUSION MATRIX TO VALIDATE MODEL

METHODS

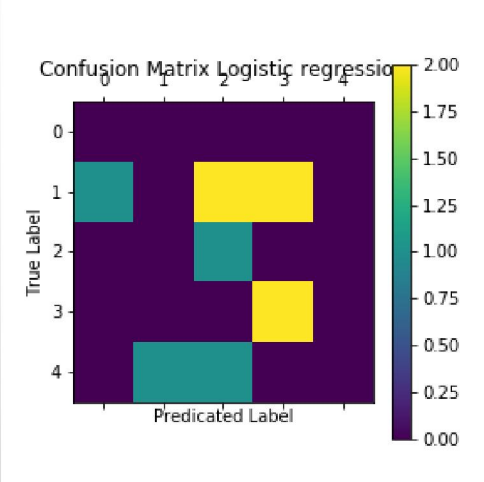
- K NEAREST NEIGHBOR
- LOGISTIC REGRESSION
- SUPPORT VECTOR MACHINE (LINEAR KERNEL)
- SUPPORT VECTOR MACHINE AFTER PCA

MODEL PERFORMANCE: THE CONFUSION MATRIX

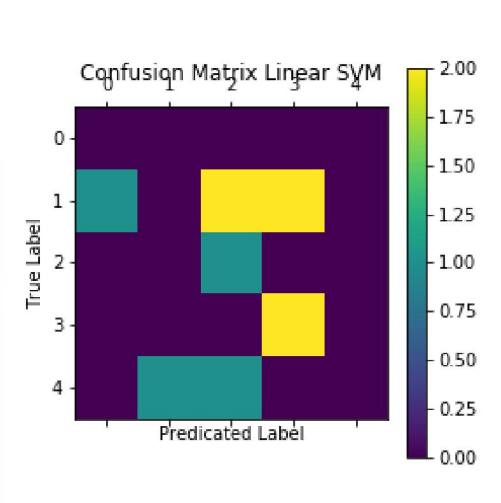
KNN



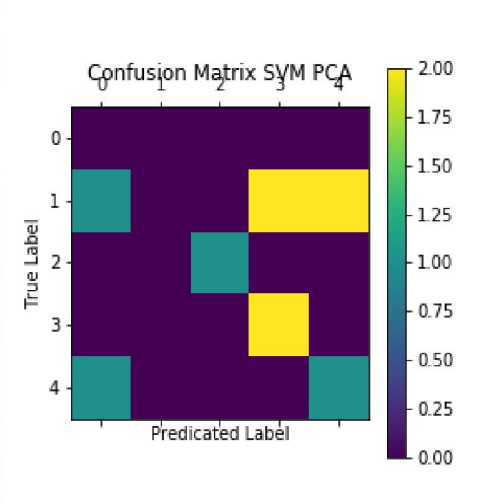
LOGISTIC REGRESSION



SVM



SVM AFTER PCA



DEXTERITY CHECK TOOL

Dexterity
CHECK

Dexterity Check

Background

Examples

PCA

All submissions

Directions: First enter your data. Use a paint software or drawing app to

Circle

Choose File No file chosen

Triangle

Choose File No file chosen

Heart

Choose File No file chosen

Star

Choose File No file chosen Upload

Uploaded files

Names: Add

Results :

Go to some interesting results

You submitted drawings of the following shapes in this order:

1.Circle

2.Triangle

3. Heart

4. Star

KNN:

Using k nearest neighbor I have identified your first,second,third, and fourth drawings as:

circle
triangle
heart
star

Dexterity
CHECK

You scored
100.0
% for dexterity using KNN!